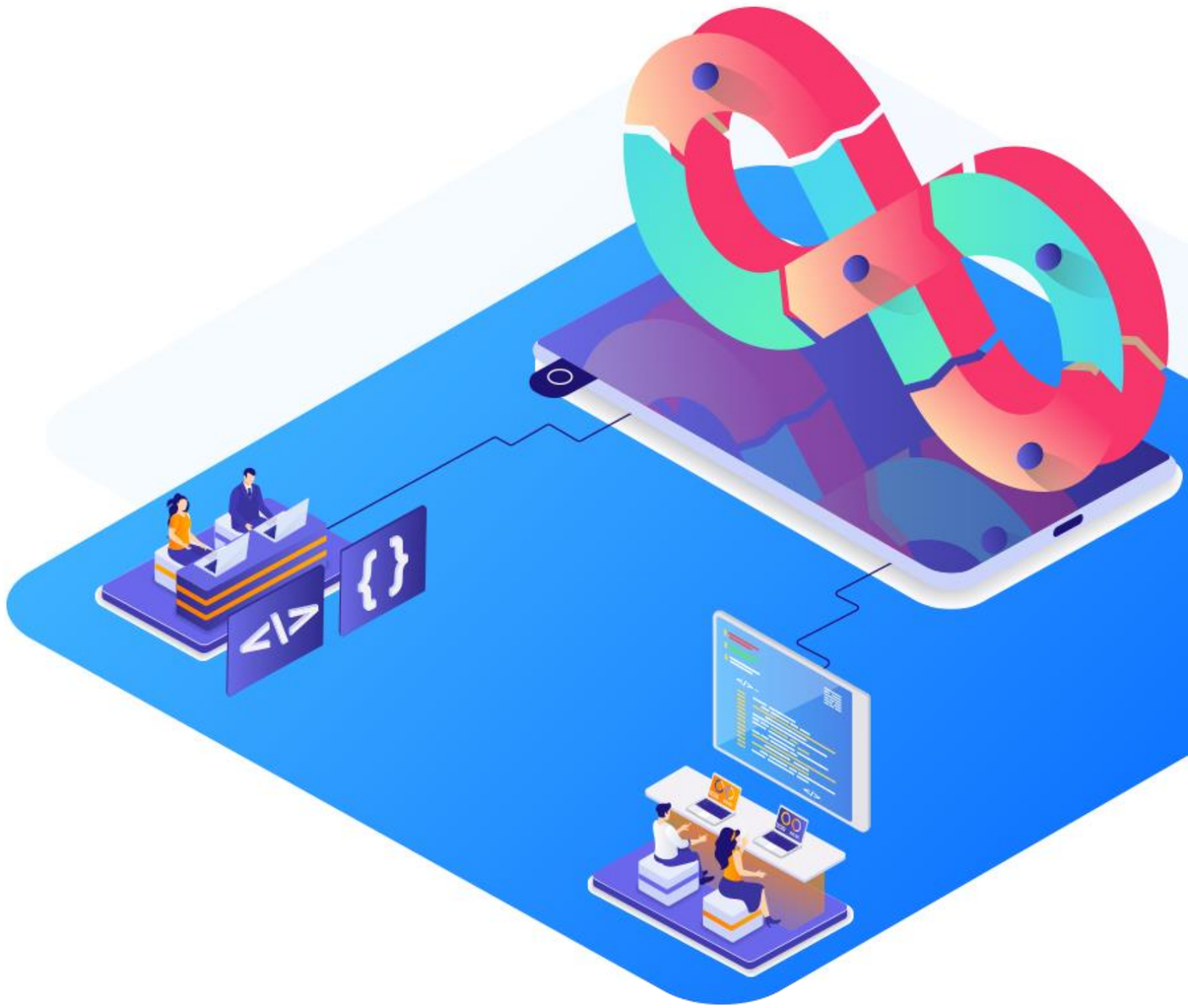
# Containerization with Docker



# Managing Docker Images and Registries

# Learning Objectives

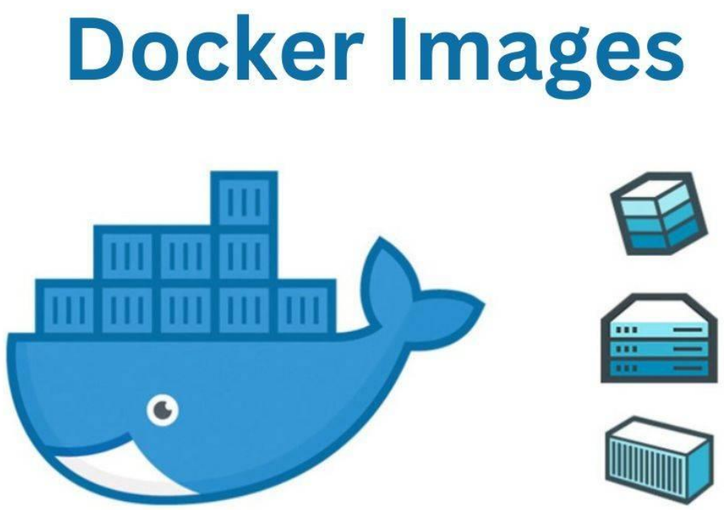
By the end of this lesson, you will be able to:

- Build Docker images using Dockerfiles, emphasizing the manipulation and optimization of image layers
- Outline the container lifecycle to allow optimization of resource allocation
- Use Docker commands to pull, push, tag, and remove images, including managing images across different environments and registries
- Set up and configure both public and private Docker registries, deploy images to these registries, and manage access and security settings

# Introduction to Docker Images and Containers

# Docker Image: Overview

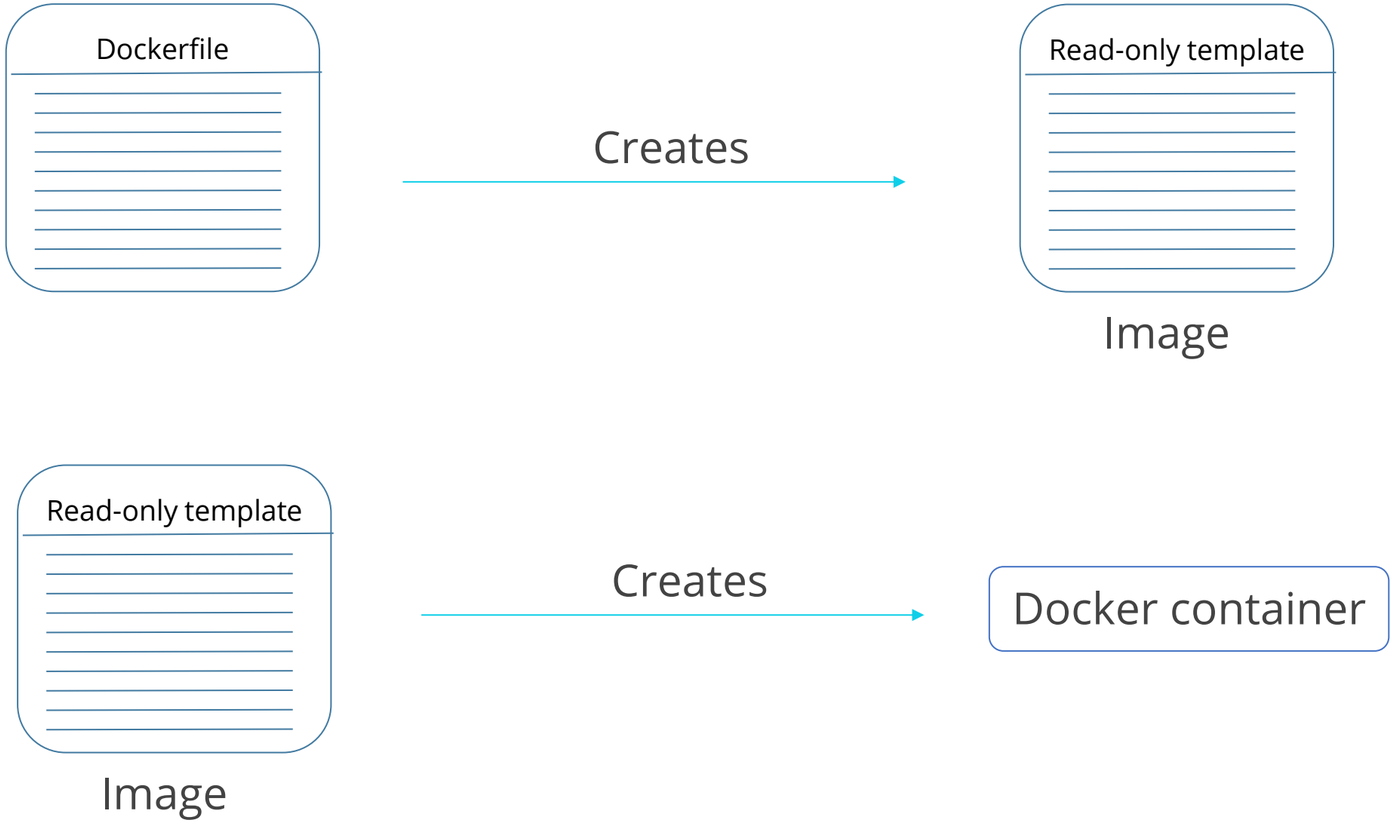
A Docker image is a file that runs programs within a Docker container. It functions as a set of guidelines like a template.




Docker images also behave as the starting point when using Docker. An image is similar to a snapshot in virtual machine (VM) environments.

# Docker Image: Overview

An image holds instructions that are required to run an application.



# Why Use Docker Images?

Docker images can be useful to:

**Maintain consistency across environments**

**Scenario:** A development team is working on a web application that must be deployed across multiple environments (development, testing, staging, and production).

**Benefit:** A Docker image ensures consistent application behavior across environments thereby reducing discrepancies and minimizing deployment risks.

# Why Use Docker Images?

## Provide scalability

**Scenario:** A business needs to scale its application to handle increased traffic during peak times (For example, Black Friday for an e-commerce site).

**Benefit:** Docker images allow the business to quickly spin up additional containers to handle the load.

## Facilitate portability

**Scenario:** A company wants to move its applications between on-premises infrastructure and the cloud without significant changes.

**Benefit:** A Docker image is portable and runs on any Docker-supported platform, simplifying application migration.

# Why Use Docker Images?

## Streamline development and deployment

**Scenario:** A software development team is adopting a CI/CD pipeline to accelerate the release of new features.

**Benefit:** A Docker image streamlines build, test, and deployment by allowing developers to work in production-like containers, catching issues early and enabling fast, reliable releases.

## Maintain isolation and security

**Scenario:** A business needs to run multiple applications on the same server without them interfering with each other.

**Benefit:** A Docker image encapsulates applications and dependencies, providing isolation that enhances security by preventing vulnerabilities in one container from affecting others.

# Parts of a Docker Image

**Base image**

The user can build this first layer entirely from scratch with the build command.

**Parent image**

This can be the first layer in a Docker image which is a reused image that serves as a foundation for all other layers.

**Layers**

Layers are added to the base image, using code that will enable it to run in a container.

# Parts of a Docker Image

**Container layer**

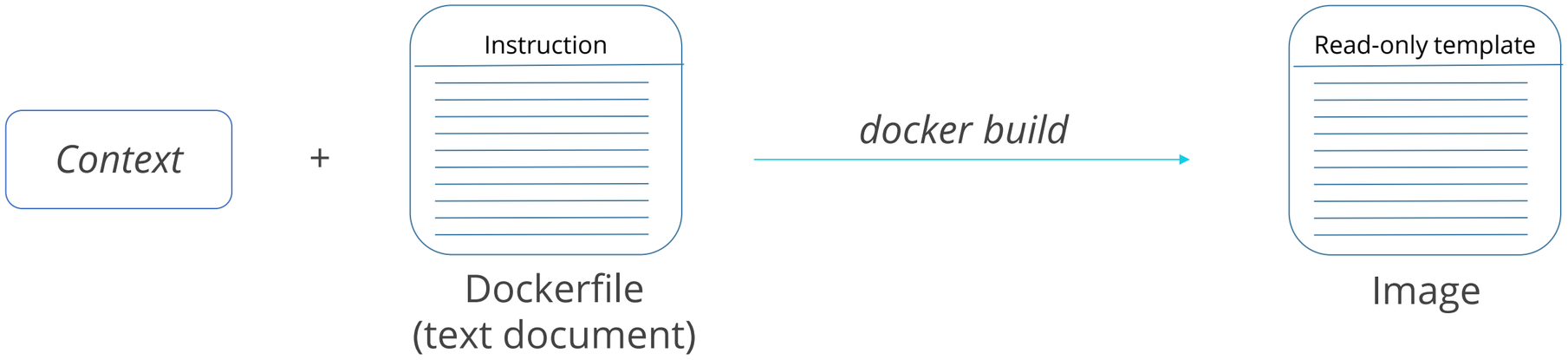
A Docker image not only creates a new container, but also a writable or container layer which is used to customize the containers.

**Docker manifest**

This part of the Docker image is an additional file which uses JSON format to describe the image.

# Dockerfile: Overview

It contains all the necessary instructions that are used to build images.




The *docker build* command creates an image from a *context* and a *Dockerfile*.

# Dockerfile: Overview

Context

Set of files

*PATH*

Directory at
local filesystem

*URL*

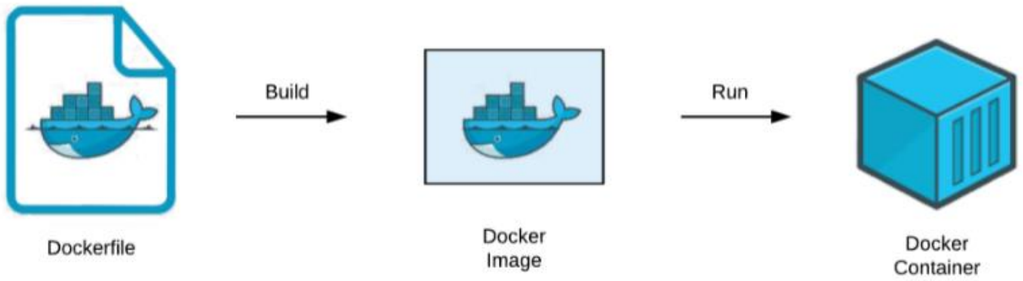
Repository location

Use of current directory in the *docker build* command

*$ docker build .*

# Docker Containers: Overview

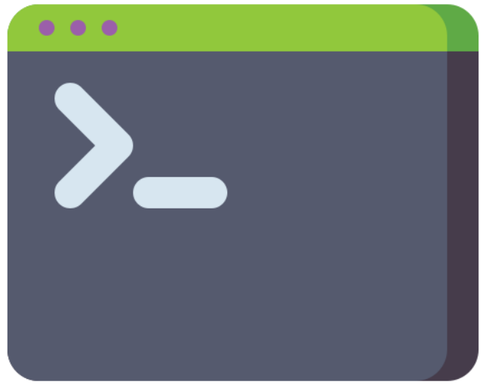
A container is a standardized software component that wraps up code and its dependencies to ensure that an application runs consistently in different computing environments.




An executable software package known as a Docker container image contains all the components required to run a program, including the code, runtime, system tools, system libraries, and settings.

# Docker Containers: Execution

A container is the same as a typical operating system process, with the exception that it is isolated and has its own file system, networking, and isolated process tree that is distinct from the host.



The **docker run** command is used to execute an image inside of a Docker container. The image name is the only parameter needed for the **docker run** command.

# Why Use Docker Containers?

Docker containers are active, stateful entities created from images. They run as isolated processes on a host machine, offering the following benefits in real-world business contexts:

## Enable microservices architecture

**Scenario:** A company is transitioning from a monolithic application to a microservices architecture, where each service needs to be independently deployed, scaled, and managed.

**Benefit:** Docker containers enable each microservice to run in its isolated environment, allowing independent deployment, scaling, and updates.

# Why Use Docker Containers?

## Standardize environments

**Scenario:** A development team is working on a complex application that requires multiple dependencies and specific configurations across different environments.

**Benefit:** Docker containers standardize the environment across development, testing, and production, ensuring that the application behaves consistently regardless of platform.

## Ensure CI/CD consistency

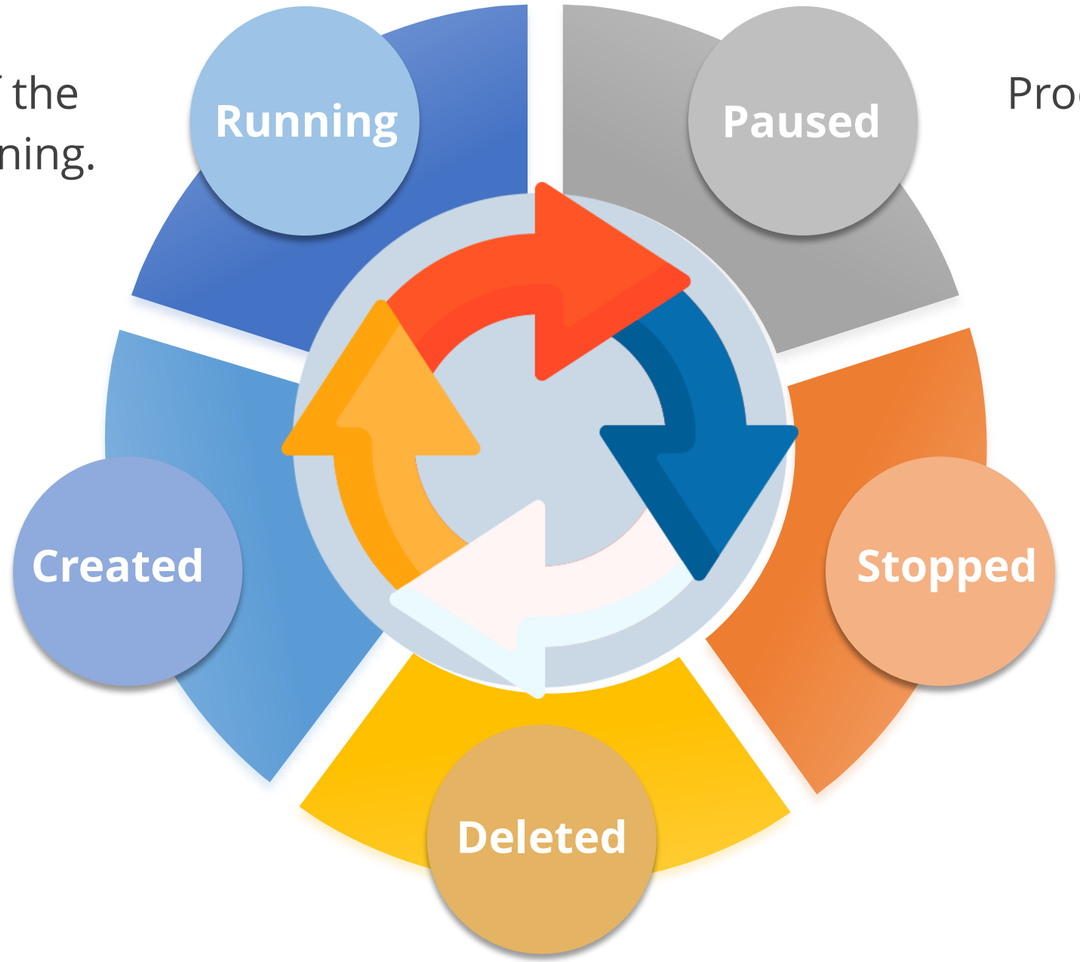
**Scenario:** A software development team implements a continuous integration/continuous deployment (CI/CD) pipeline to accelerate their development process.

**Benefit:** Docker containers play a critical role in CI/CD by providing consistent, reproducible environments for automated testing, staging, and production.

# Container Lifecycle

Here are the main phases in the lifecycle of a Docker container:

The processes of the container are running.




Processes have been paused for some time.

The container is created but not started.

The process has been stopped.

The container is deleted, including all its processes.

# Container Lifecycle

To create the new Docker container with an image, use the given command:

```
$ docker create  --name <container-name> <image-name>
```

# Container Lifecycle

To start a newly created or stopped Docker container, use the **docker start** command.

```
$ docker start  <container-name>
```

Example:

```
$ docker start docker-container-2022
```

# Container Lifecycle

The **docker run** command performs the tasks of both the **docker create** and **docker start** commands.

```
$ docker run -it --name  <container-name>
```

Example:

```
$ docker run -it --name docker-container-2022
```

# Container Lifecycle

The **docker pause** command is used to pause a running container by specifying its name.

```
$ docker pause  <container-name>
```

Example:

```
$ docker pause docker-container-2022
```

# Container Lifecycle

Syntax:

```
$ docker stop  <container-name>
```

Example:

```
$ docker stop docker-container-2022
```

# Container Lifecycle

To delete all the containers, use the given command:

```
$ docker rm $(docker container ps -a)
```

Example:

```
$ docker rm $(docker container ps -a)
a2f73432da1c
dda9c86d423e
3b326aa26db6
2668cf8c9aab
```

# Container Lifecycle

When using the **docker stop** command, all processes running in the designated container are stopped.

A stopped container can be transitioned into a running state.

# Container Lifecycle

Before deleting a Docker container, it is necessary to stop all running processes inside it.

It is recommended to pause the container before making any modifications to avoid encountering errors.

# Assisted Practice

**Demonstrating lifecycle of containers** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to demonstrate the lifecycle of containers for efficient management and optimization of Docker container orchestration.

**Outcome:**

By completing this demo, you will be able to effectively manage the lifecycle of Docker containers, ensuring efficient container orchestration and optimization.

**Note**: Refer to the demo document for detailed steps
01_Demonstrating_Lifecycle_of_Containers

# Assisted Practice: Guidelines

Steps to be followed:

1. Demonstrate Docker container lifecycle management

ASSISTED PRACTICE

# Assisted Practice

**Creating a Docker Image** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to create a Docker image using a Docker file, which installs and configures a Nginx web server with a custom welcome message.

**Outcome:**

By completing this demo, you will be able to create a Docker image containing a Nginx web server with a custom welcome message using a Dockerfile. This image can then be used to spin up containers with the configured web server environment.

**Note**: Refer to the demo document for detailed steps
02_Creating_a_Docker_Image

# Assisted Practice: Guidelines

Steps to be followed:

1. Create a Docker image using the Docker file

ASSISTED PRACTICE

# Quick Check

You are working on a DevOps project and want to debug specific application components. Which command will you use to pause a running container by specifying its name?

A. $ docker hold execution myContain

B. $ docker pause myContain

C. $ docker break –it --name myContain

D. $ docker kill myContain

# Image Management

# Why Is It Necessary?

**Efficient resource management**

Proper image management minimizes storage space by avoiding redundant images, leading to more efficient resource use and cost savings

**Consistency and reliability**

Managing images ensures that applications run consistently across different environments, reducing deployment issues and improving reliability

**Security and compliance**

Effective image management helps maintain security by using up-to-date images and scanning for vulnerabilities, minimizing the risk of breaches

**Scalability and automation**

Enable faster deployments and smooth scalability in automated environments like Kubernetes, ensuring seamless operations at scale

# Image Creation Techniques

## Interactive method

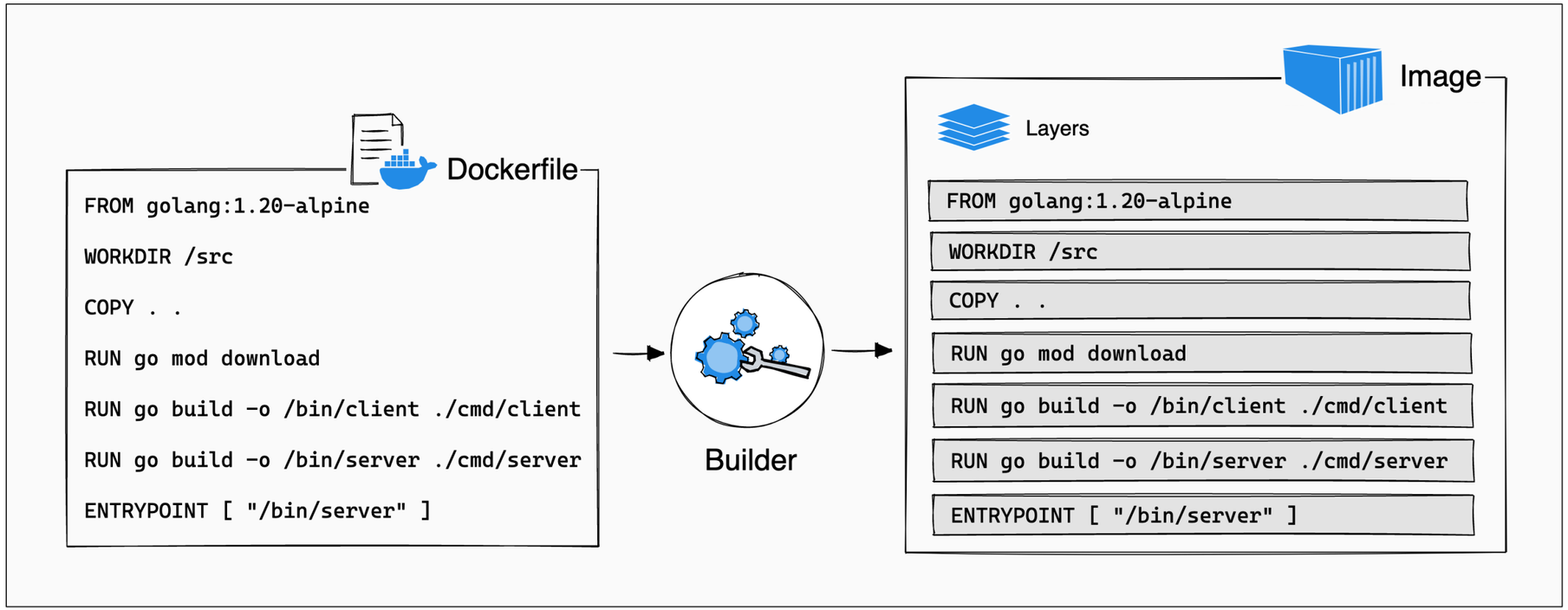
- This is the easiest way to create Docker images.
- With this method, users run a container from an existing Docker image and manually make any needed changes to the environment before saving the image.
- **docker build -t <image_name>:<tag>** is the command to create a docker image through CLI.

## Dockerfile method

- This process is more difficult and time-consuming, but it does well in continuous delivery environments.
- This approach requires making a plain text Dockerfile. The Dockerfile makes the specifications for creating an image.
- Dockerfile contains all necessary dependencies defined in it to build a docker image

# Layers of Docker Image

A Docker image is composed of multiple layers stacked on top of each other. The following diagram illustrates how a Dockerfile translates into a stack of layers in a container image:




*Source: https://docs.docker.com/build/guide/layers/*

# Layers of Docker Image

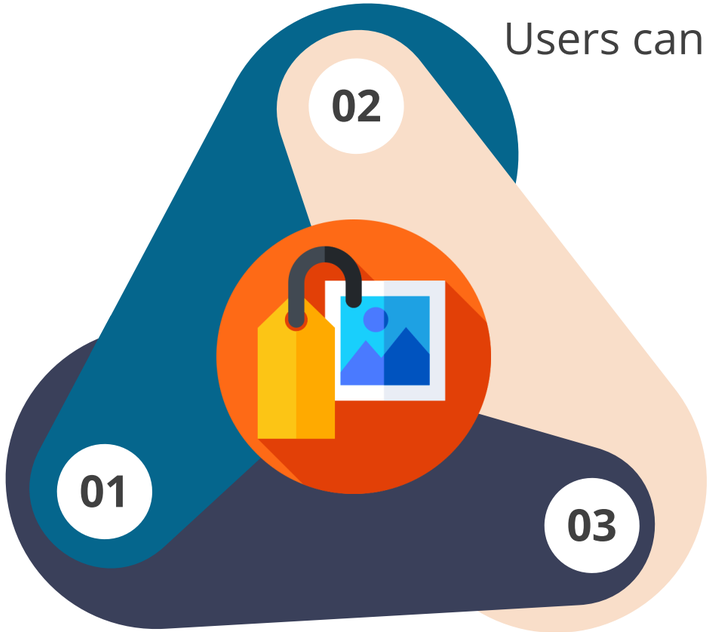
Building layers consists of the following four major components:

| | |
|---|---|
| **Dockerfile** | It is a file that contains the commands or instructions to build layers. |
| **Imagedb** | It stores images in registries like Docker Hub. |

# Layers of Docker Image

| | |
|---|---|
| **Layerdb** | It stores the current working state of layers as they are built at each command from Dockerfile. |
| **Caches** | It stores objects or files for images so they can be cached to speed up retrieval time. |

# How to Tag an Image?




Users can tag Docker images based on their preferences.

Identifying a specific image becomes difficult when there are multiple images.

An image name comprises of name components separated by slashes.

# Tagging an Image

**Tag by ID**

**Tag by name**

**Tag by name and tag**

**Tag for a private repository**

Users employ it to tag a local image with ID *0e93876876* and associate it with the spring repository using version 1.0.

```
docker tag 0e93876876 spring/myapp:version1.0
```

# Tagging an Image

- Tag by ID
- **Tag by name**
- Tag by name and tag
- Tag for a private repository

It is used to tag a local image with the name *myapp* into the spring repository with version1.0:

```
docker tag myapp spring/myapp:version1.0
```

# Tagging an Image

- Tag by ID
- Tag by name
- **Tag by name and tag**
- Tag for a private repository

It is used to tag a local image with the name *myapp* and tag *test* into the spring repository with version1.0.test:

```
docker tag httpd:test spring/myapp:version1.0.test
```

# Tagging an Image

Tag by ID

Tag by name

Tag by name and tag

Tag for a private repository

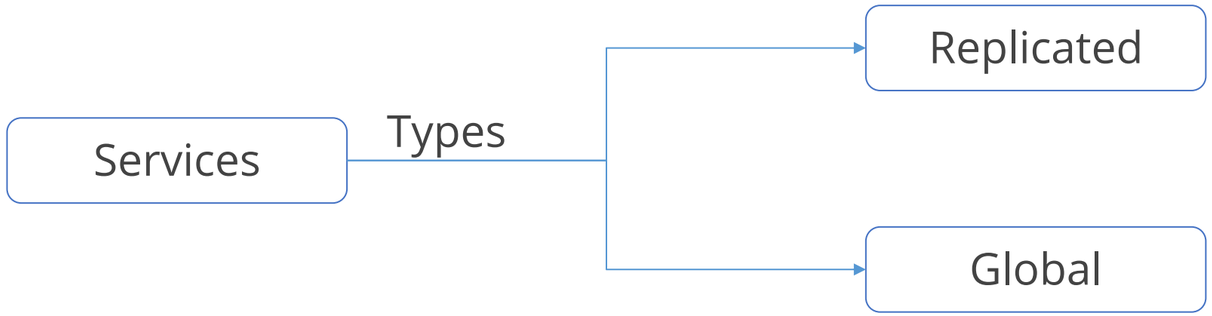
It facilitates pushing an image to a private registry rather than the central Docker registry.

```
docker tag 0e93876876
myregistryhost:9090/spring/myapp:version1.0
```

# Services and Tasks: Overview

Services are part of the image that perform specific tasks, allowing you to scale containers across multiple Docker Daemons. These Daemons work together in a network with various manager and worker nodes.

Each member of a network is a Docker Daemon that communicates with other Daemons using the Docker API.




A service allows you to define the desired state, such as the number of replicas of the service that must be available at any given time. By default, the service is load-balanced across all worker nodes.

# Docker Commit

This command commits a container's changes or settings to a new image. By default, all processes are paused during the commit process, which helps reduce data corruption.

Committing a container:

```
$ docker ps
$ docker commit ContainerID repository:tag
$ docker images
```

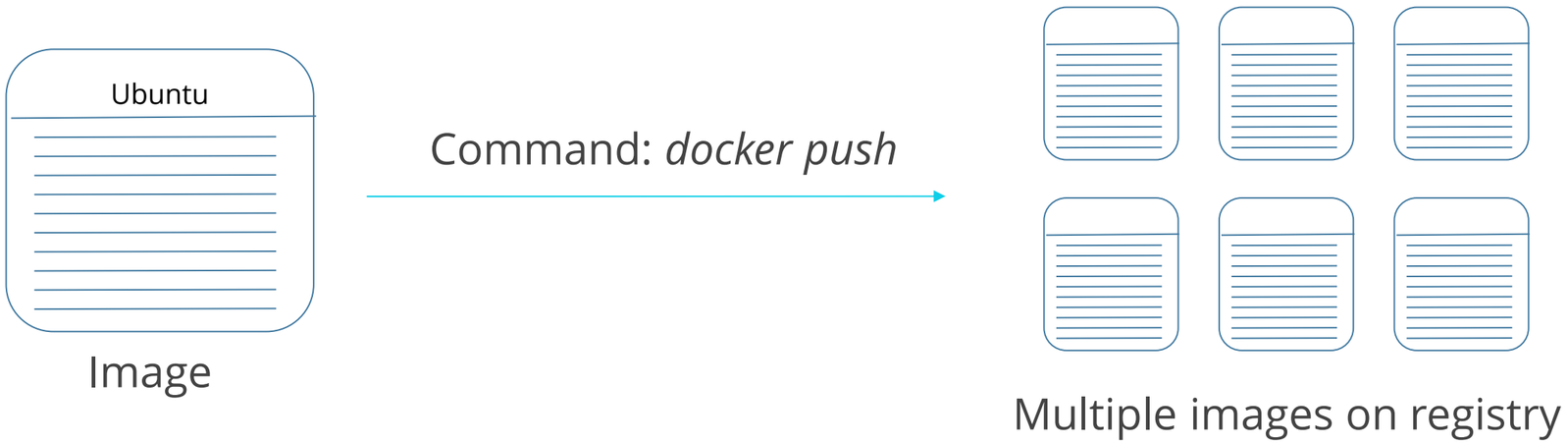
While committing, the **--change** option is used to make changes in Dockerfile instructions such as CMD, ENTRYPOINT, ENV, EXPOSE, LABEL, ONBUILD, USER, VOLUME, and WORKDIR.

# Docker Push Command

The **docker push** command is used to upload or push Docker images from a local environment to a Docker registry, such as Docker Hub, Google Container Registry, Amazon ECR, or a private registry.

**Syntax:**

```
docker push [OPTIONS] NAME[:TAG]
```

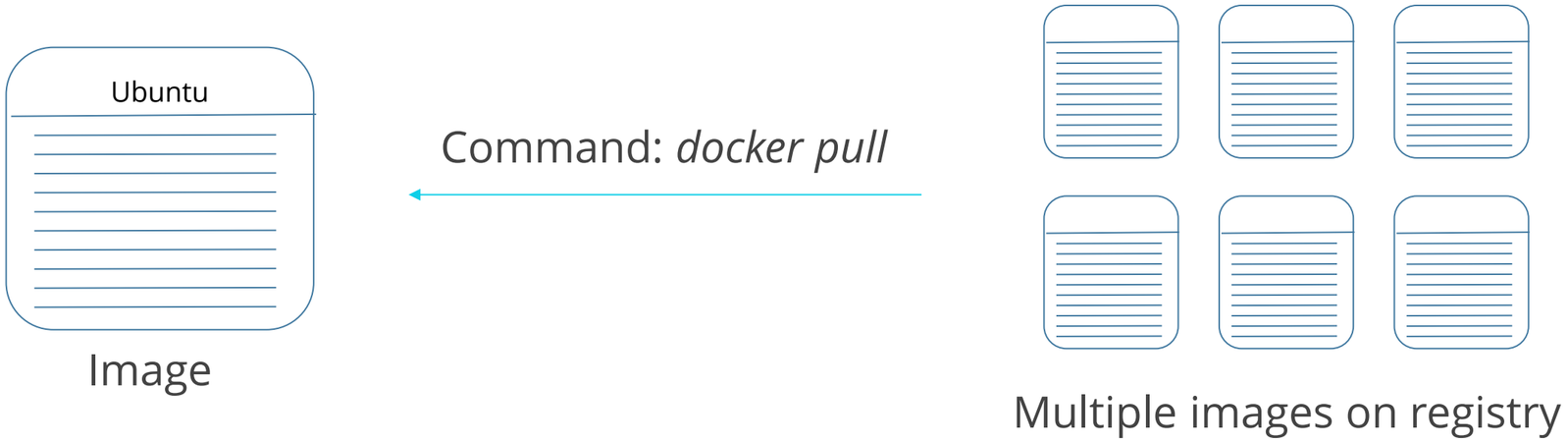


# Docker Pull Command

The **docker pull** command downloads Docker images from a Docker registry to the local machine. It is generally needed to set up environments using the pre-built images.

**Syntax:**

```
docker pull [OPTIONS] NAME[:TAG]
```



# Assisted Practice

**Tagging an image** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to demonstrate the process of tagging Docker images using various methods for easier management and distribution.

**Outcome:**

By completing this demo, you will be able to tag Docker images with different versions and repository names, facilitating easier management and distribution of Docker images.

**Note**: Refer to the demo document for detailed steps
03_Tagging_an_Image

# Assisted Practice: Guidelines

Steps to be followed:

1. Pull a Docker image
2. Tag an image referenced by name
3. Tag an image for a private repository

ASSISTED PRACTICE

# Assisted Practice

**Displaying layers of a Docker image** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to display the layered structure of Docker images for providing insight into the hierarchical arrangement of image layers within Docker's architecture.

**Outcome:**

By completing this demo, you will be able to create a Docker image with multiple layers and display the layered structure, providing insight into the hierarchical arrangement of image layers within Docker's architecture.

**Note**: Refer to the demo document for detailed steps
04_Displaying_Layers_of_a_Docker_Image

# Assisted Practice: Guidelines

Steps to be followed:

1. Create and display the layers of a Docker image

ASSISTED PRACTICE

# Quick Check

As a DevOps engineer, which image layers will you modify to store the current working state of layers as they are built at each command from Dockerfile?

A. Layerdb

B. Caches

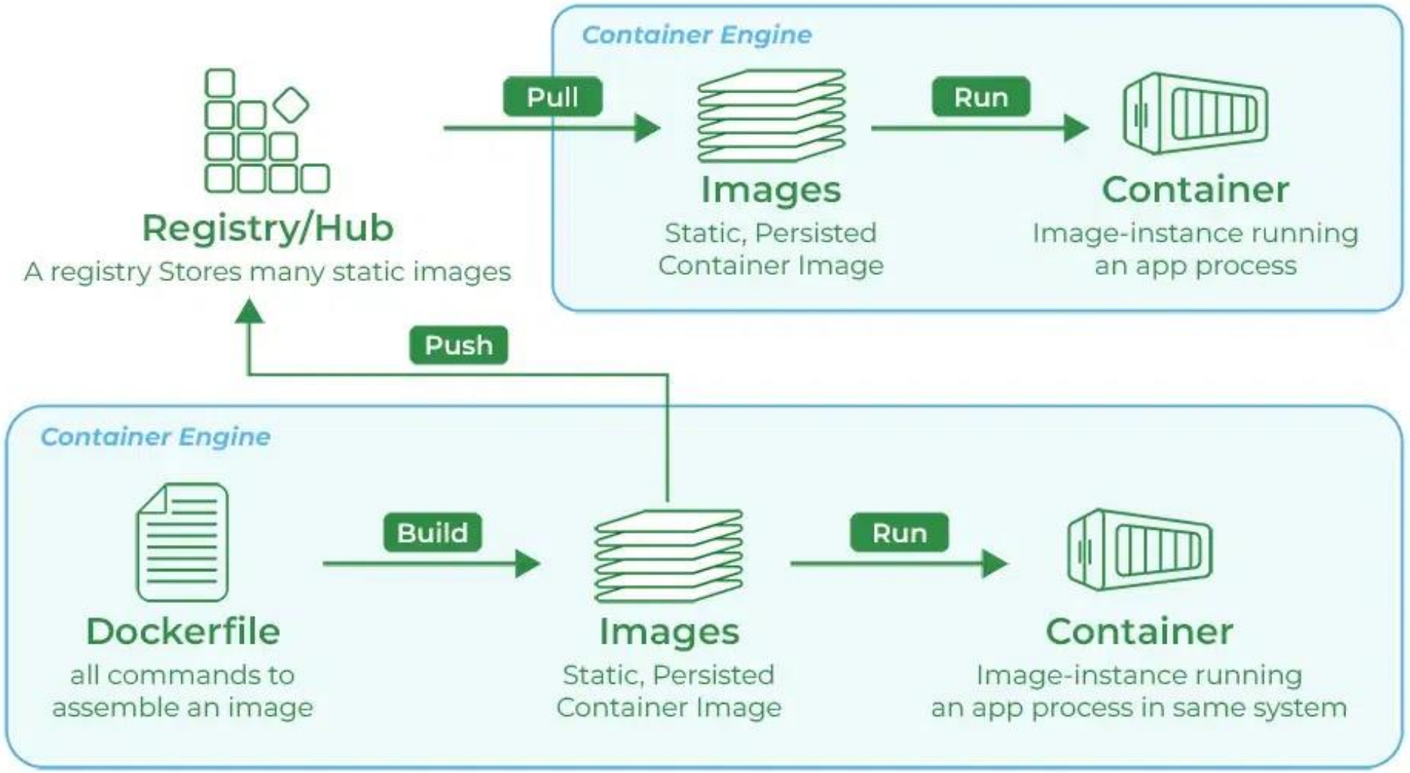
C. Imagedb

D. Dockerfile

# Introduction to Docker Registry

# Docker Registry: Overview

A Docker registry is a system for storing and distributing Docker images with specific names. It is separated into Docker repositories, each holding all image modifications.




*Source: https://www.geeksforgeeks.org/what-is-docker-registry/*

# Docker Registry: Features

01 Docker registries provide a centralized platform for storing and distributing container images efficiently.

02 Users can maintain multiple versions of the same image within the registry, allowing for version control and image management.

03 Users can set up private registries for storing sensitive images securely, enabling controlled access within an organization.

# Why Use Docker Registries?

Docker registries offer the following benefits in real-world business contexts:

## Centralized image management

**Scenario:** A software development team is working on multiple microservices that need to be consistently updated and deployed across various environments.

**Benefit:** A Docker registry serves as a central hub for storing and managing Docker images, ensuring that all team members have access to the latest version of each microservice.

# Why Use Docker Registries?

## Secure image distribution

**Scenario:** A financial institution needs to ensure that only approved and secure Docker images are used in production.

**Benefit:** Docker registries provide a controlled environment for storing and distributing Docker images, providing access control mechanisms.

## Version control and rollbacks

**Scenario:** After deploying a new version of an application, critical issues are discovered, requiring a quick rollback to the previous stable version.

**Benefit:** Docker registries maintain a history of all image versions, making it easy to roll back to a previous version when needed.

# Docker Registry: Commands

| Description | Commands |
|---|---|
| Starting your registry | docker run -d -p 5000:5000 --restart=always --name registry registry:2 |
| Pulling images from the hub | docker pull ubuntu:latest |
| Tagging an image and point to registry | docker image tag ubuntu:latest localhost:5000/gfg-image |
| Pushing the image | docker push localhost:5000/gfg-image |
| Pulling the image back | docker pull localhost:5000/gfg-image |
| Stop the registry | docker container stop registry |

# Assisted Practice

**Deploying and Configuring a Registry** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to configure a local Docker registry for efficient storage and transfer of container images, facilitating streamlined image management within the development environment.

**Outcome:**

By completing this demo, you will be able to deploy and configure a local Docker registry, including pushing, pulling, and managing Docker images within the registry.

**Note**: Refer to the demo document for detailed steps
05_Deploying_and_Configuring_a_Registry

# Assisted Practice: Guidelines

Steps to be followed:

1. Run a local registry

ASSISTED PRACTICE

# Assisted Practice

**Pushing an Image to Registry** **Duration: 10 Min.**

**Problem statement:**

You have been assigned a task to copy an image from Docker Hub to your local Docker registry.

**Outcome:**

By completing this demo, you will be able to copy the Nginx image from Docker Hub to your local registry, demonstrating the process of transferring images for efficient local image management.

**Note**: Refer to the demo document for detailed steps
06_Pushing_an_Image_to_Registry

# Assisted Practice: Guidelines

Steps to be followed:

1. Copy an image from the docker hub to your registry

ASSISTED PRACTICE

# Quick Check

You are tasked with pulling a specific version of an image from a private Docker registry hosted at myregistry.local:5000. The image is named myapp, and you need the version tagged as 3.1.4. Which of the following commands should you use to correctly pull the image from the registry?

A. docker pull myregistry.local:5000/myapp

B. docker pull myregistry.local:5000/myapp:latest

C. docker pull myregistry.local:5000/myapp:3.1.4

D. docker image pull myregistry.local:5000/myapp

# Key Takeaways

- A Docker image is a file that runs programs within a Docker container. It functions as a set of guidelines like a template.
- A container is the same as a typical operating system process, except that it is isolated and has its own file system, networking, and isolated process tree distinct from the host.
- A container is a standardized software component that wraps up code and its dependencies to ensure that an application runs consistently in different computing environments.
- A Docker registry is a system for storing and distributing Docker images with specific names.
- Docker provides pruning functionality to help manage unused images efficiently and improve system performance.

# Dockerizing a Java Program and Setting up a Secure Local Registry

**Duration: 25 min.**

**Project Agenda**: To establish a secure local registry and dockerize a Java program for efficient and consistent deployment across various environments

**Description:** Your company is experiencing a need to streamline deployment processes and ensure consistency across software environments. To address this, you are undertaking a project to dockerize a Java program and set up a local registry with security measures that enhance scalability and maintain deployment consistency.

LESSON-END PROJECT

# Dockerizing a Java Program and Setting up a Secure Local Registry

**Duration: 25 min.**

**Perform the following:**

1. Write a Java program and create a Dockerfile
2. Create a local Docker registry and run it with an htpasswd file
3. Build and tag the Docker image
4. Log in to the secured registry and push the Docker image to it

**Expected deliverables:** Dockerized Java program and a secure local registry

LESSON-END PROJECT

# Thank You